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Roll No.: C-04

Software Proficiency Program II (Python Programming)

Experiment no.:5

1. Write a program to check whether a given number is positive, negative, or zero.

```
num = float(input("Enter a number: "))
```

```
if num > 0:
```

```
    print("The number is positive.")
```

```
elif num < 0:
```

```
    print("The number is negative.")
```

```
else:
```

```
    print("The number is zero.")
```

```
Enter a number: 2
The number is positive.
```

2. Write a program to check whether a given number is even or odd.

```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0:
```

```
    print("The number is even.")
```

```
else:
```

```
    print("The number is odd.")
```

```
Enter a number: 4
The number is even.
```

3. Write a program to check whether a given character is a vowel or consonant.

```
ch = input("Enter a character: ")
```

```
if ch.lower() in ['a', 'e', 'i', 'o', 'u']:
```

```
    print("The character is a vowel.")
```

```
else:
```

```
    print("The character is a consonant.")
```

```
Enter a character: M
The character is a consonant.
```

4. Write a program to check whether a given number is divisible by 5 and 11.

```
num = int(input("Enter a number: "))
```

```
if num % 5 == 0 and num % 11 == 0:
```

```
    print("The number is divisible by both 5 and 11.")
```

```
else:
```

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```
print("The number is not divisible by both 5 and 11.")
```

```
Enter a number: 54
```

```
The number is not divisible by both 5 and 11.
```

5. Write a program to check whether a given triangle is valid (sum of any two sides greater

```
a = float(input("Enter first side: "))
```

```
b = float(input("Enter second side: "))
```

```
c = float(input("Enter third side: "))
```

```
if a + b > c and a + c > b and b + c > a:
```

```
    print("The triangle is valid.")
```

```
else:
```

```
    print("The triangle is not valid.")
```

```
Enter first side: 4
```

```
Enter second side: 5
```

```
Enter third side: 6
```

```
The triangle is valid.
```

6. Write a program to check whether a person is eligible to vote (age $\geq$ 18).

```
age = int(input("Enter your age: "))
```

```
if age >= 18:
```

```
    print("You are eligible to vote.")
```

```
else:
```

```
    print("You are not eligible to vote.")
```

```
Enter your age: 20
```

```
You are eligible to vote.
```

7. Write a program to calculate the grade of a student based on marks (A: $\geq$ 90, B: 75–89, C: 50–74, F:

```
marks = float(input("Enter your marks: "))
```

```
if marks >= 90:
```

```
    print("Grade: A")
```

```
elif marks >= 75:
```

```
    print("Grade: B")
```

```
elif marks >= 50:
```

```
    print("Grade: C")
```

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else:

```
print("Grade: F")
```

```
Enter your marks: 85
```

```
Grade: B
```

8. Write a program to check whether a given number is a multiple of 3 or 7.

```
num = int(input("Enter a number: "))
```

```
if num % 3 == 0 or num % 7 == 0:
```

```
    print("The number is a multiple of 3 or 7.")
```

```
else:
```

```
    print("The number is not a multiple of 3 or 7.")
```

```
Enter a number: 28
```

```
The number is a multiple of 3 or 7.
```

9. Write a program to check whether a given year is a century year (divisible by 100).

```
year = int(input("Enter a year: "))
```

```
if year % 100 == 0:
```

```
    print("It is a century year.")
```

```
else:
```

```
    print("It is not a century year.")
```

```
Enter a year: 2025
```

```
It is not a century year.
```

10. Write a program to check whether a given temperature is hot, warm, or cold (hot: >35°C, warm: 20–35°C, cold: <20°C). `def check_temperature():`

```
temp = float(input("Enter temperature in °C: "))
```

```
if temp > 35:
```

```
    print("It's hot.")
```

```
elif temp >= 20:
```

```
    print("It's warm.")
```

```
else:
```

```
    print("It's cold.")
```

```
Enter temperature in °C: 96
```

```
It's hot.
```